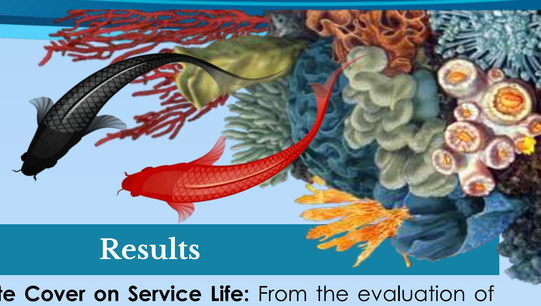
SIGNIFICANCE OF SERVICE LIFE BASED CONCRETE MIX DESIGN IN MARINE ENVIRONMENT

Group-6

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Abstract

The durability of reinforced concrete structures in marine environments is often not ensured by traditional mix design methods that prioritize strength. The impermeability of concrete, as measured by its chloride diffusion coefficient, can affect the corrosion initiation of reinforcement. Mix design practices that focus solely on strength may not ensure reduced diffusion coefficients. The choice of cement type and concrete cover can significantly impact the service life of RC elements in marine conditions.

Introduction

- Strength-based design lacks in safe-guarding durability of RC structures.
- Concrete serviceability relating to permeability is rarely addressed.
- In this study, the current concrete mix was evaluated for typical cover size.
- Mix design followed a volumetric mixing ratio of 1:1.5:3 and w/c ratio of 0.45~0.55 yielding moderate compressive strength (20MPa to 38MPa).
- Irrespective of Cover values and exposure conditions, PCC ensures better serviceability than OPC .

Methods and Materials

01 Sample Selection

A total of 11 local concrete mix variations were prepared.

02 Standards for Sampling

.All concrete samples were tested for their diffusion coefficient values as per the migration test specification **NT BUILD 492 (1999).**

03 Sample Preparation

then the sample was placed within the rubber sleeve of the setup, as shown in Figure 1



Figure 1

04 Sample Testing

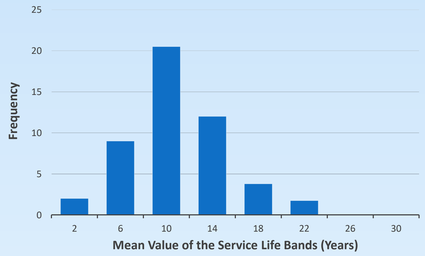
After the test completion, the sample was split into two halves and the diffusion coefficient was determined from the penetration depth as stated in NT BUILD 492 (1999).

Results

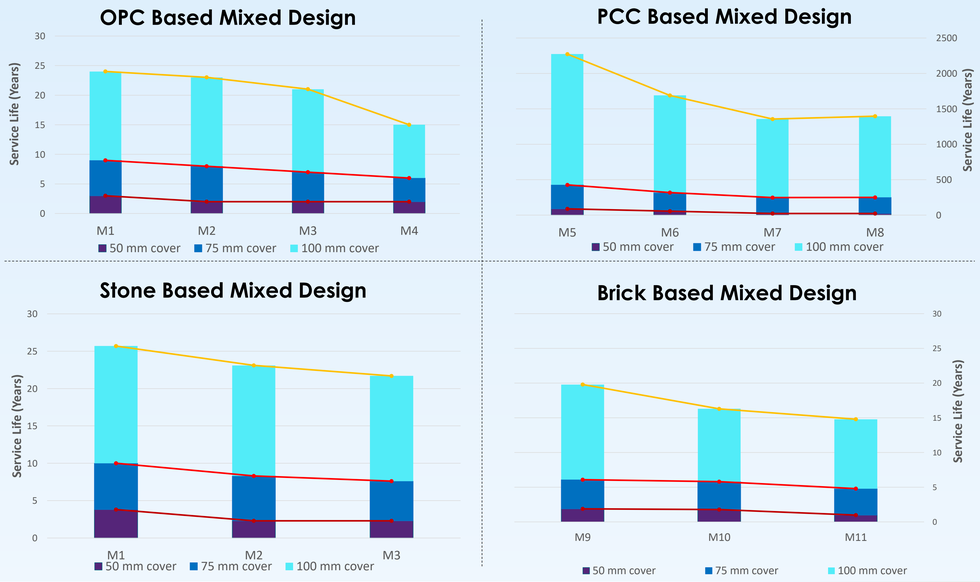
1. Diffusion Coefficients: The experimented diffusion coefficients of all the mixes considered were evaluated as per the specifications of NT BUILD 492 (1999) is shown in the following table:

Mix Type	Diffusion Coefficient (m2/s)	Mix Type	Diffusion Coefficient (m /s)
M1	2.3778E-11	M7	1.89844E-11
M2	2.415E-11	M8	1.80604E-11
M3	2.5E-11	M9	2.477E-11
M4	3E-11	M10	2.5E-11
M5	1.57487E-11	M11	2.7E-11
M6	1.7511E-11		

2. Probabilistic Evaluation of Service Life: The experimented diffusion coefficients of all the mixes considered were evaluated as per the specifications of NT BUILD 492 (1999) is shown in the following table:

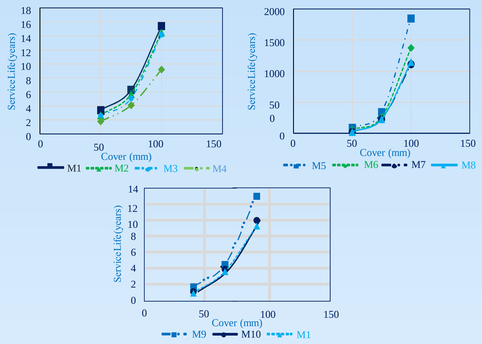


3. Effect of Mix Variations on Service Life: Variations in mean values of the service life distributions of different mix types for typically used concrete cover of 50mm, 75mm and 100mm are presented in the form of bar charts



Results

4. Effect of Concrete Cover on Service Life: From the evaluation of service life of different common local concrete mixes under extreme marine exposure, the assessment shows that for concrete covers of 50 mm, 75 mm & 100 mm the service life changes about exponentially in the following figures.



Conclusion

- Concrete mixes with PPC and stone aggregate has higher corrosion resistance and long durability compared to OPC and brick.
- For long serviceability of the structure, concrete cover along with concrete mixes must be considered.
- OPC blends needs higher concrete cover than PPC blends.

References

1. Manzur, Tanvir & Hasan, Md & Baten, Bayezid & Torsha, Tafannum & Khan, Md Fahim & Hossain, K.. (2019). SIGNIFICANCE OF SERVICE LIFE BASED CONCRETE MIX DESIGN IN MARINE ENVIRONMENT.

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